BabyTalk — System Architecture

Updated: 2026-07-29
Workspace branch: `cursor/iou-improvements` (1 commit ahead of `main`: VTC-first segmentation)
Also documents: uncommitted worktree changes on that branch (review UI theme, supervised launcher, analysis sweeps)
Related docs: phone waveform deep dive in `ARCHITECTURE_DEEP_DIVE.md`; Mac ops detail in `../tools/README.md`; product phasing in `PRODUCT_ROADMAP.md`

This is the maintainable system map for **phone capture → Mac review → tags/library**. Numbers below are from checked-in or worktree analysis JSON under `tools/analysis/out/` — not invented.

Status legend

Marker	Meaning
main	On `main` today (tip ≈ merge-base `80063ee` — DJW resegment + ECAPA ML path + Review Browser)
branch	Committed on `cursor/iou-improvements` only (`ad849e3` — VTC / `vtc-first`)
WIP	Present in the working tree but **not committed** (experimental / not merged)

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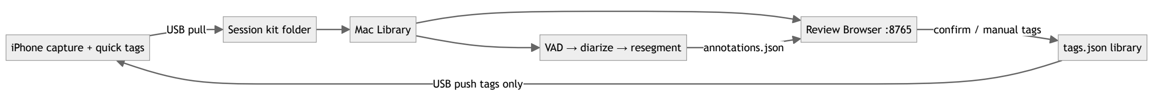
1. Product purpose

BabyTalk is a personal research instrument: capture a child's vocalizations on phone, keep durable session kits, and grow a structured tag library (words, phonetics, categories, speakers) for later modeling.

Layer	Job
Phone (React Native)	Record ALAC/WAV sessions, live + Path-B waveforms, quick tags, export session kits, USB File Sharing
Mac Review Browser	Listen, drag-span tag, confirm/dismiss ML candidates, cluster similar sounds, USB sync
Mac ML tools	Propose tag-sized candidates (`annotations.json`); never overwrite confirmed human labels

Near-term thesis: the **labeling flywheel** (propose → confirm → better proposals) runs on the Mac; the phone stays the capture + quick-tag edge. Google Drive / cloud ML are out of scope for now.



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2. What's on `main` vs this branch

On `main` today

- RN app: record / playback / Path B peaks / session naming / category+speaker+word+phonetic tagging
- Session kits under `~/Documents/BabyTalk/Library/` + USB sync (`tools/iphone_sync.py`)
- Review Browser (`tools/review_server.py`): waveform tagging, ML candidates, Clustering tab, Sync button
- ML SoTA path: **energy VAD + speechlike gate → ECAPA diarization → pause-split → DJW syllable resegment → annotations**
- Supporting modules: `vad_segments.py`, `speechlike.py`, `diarize.py`, `resegment.py`, `cluster_sounds.py`, `asr_suggest.py`
- Benchmark harness (partial): `tools/analysis/ml_delta.py`

On branch `cursor/iou-improvements` (committed)

- `tools/vtc.py` — LAAC-LSCP Voice Type Classifier wrapper
- `--segmentation vtc-first` and `--diarization vtc` in `vad_segments.py`
- Review / README wiring for VTC backends

- Extended `ml_delta.py` for fresh runs + segmentation flags

Verdict from benchmarks (below): keep **ECAPA** as default stage-2; do **not** replace it with VTC on these phone-mic kits.

WIP (worktree, uncommitted)

- Supervised launcher `tools/run_review_server.sh` (survives agent shell deaths)
- Dark theme with **Auto = evening/night** (18:00–06:00) + Light/Dark cycle
- Tags list **scroll viewport** (`.tag-rows` ≈ 20 compact rows, `overflow-y: auto`)
- Analysis: `iou_sweep.py` , `role_delta.py` , outputs under `tools/analysis/out/`
- `resegment.py` parameter plumbing for word-split knobs (defaults unchanged; enables sweeps)

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3. Mac Review Browser

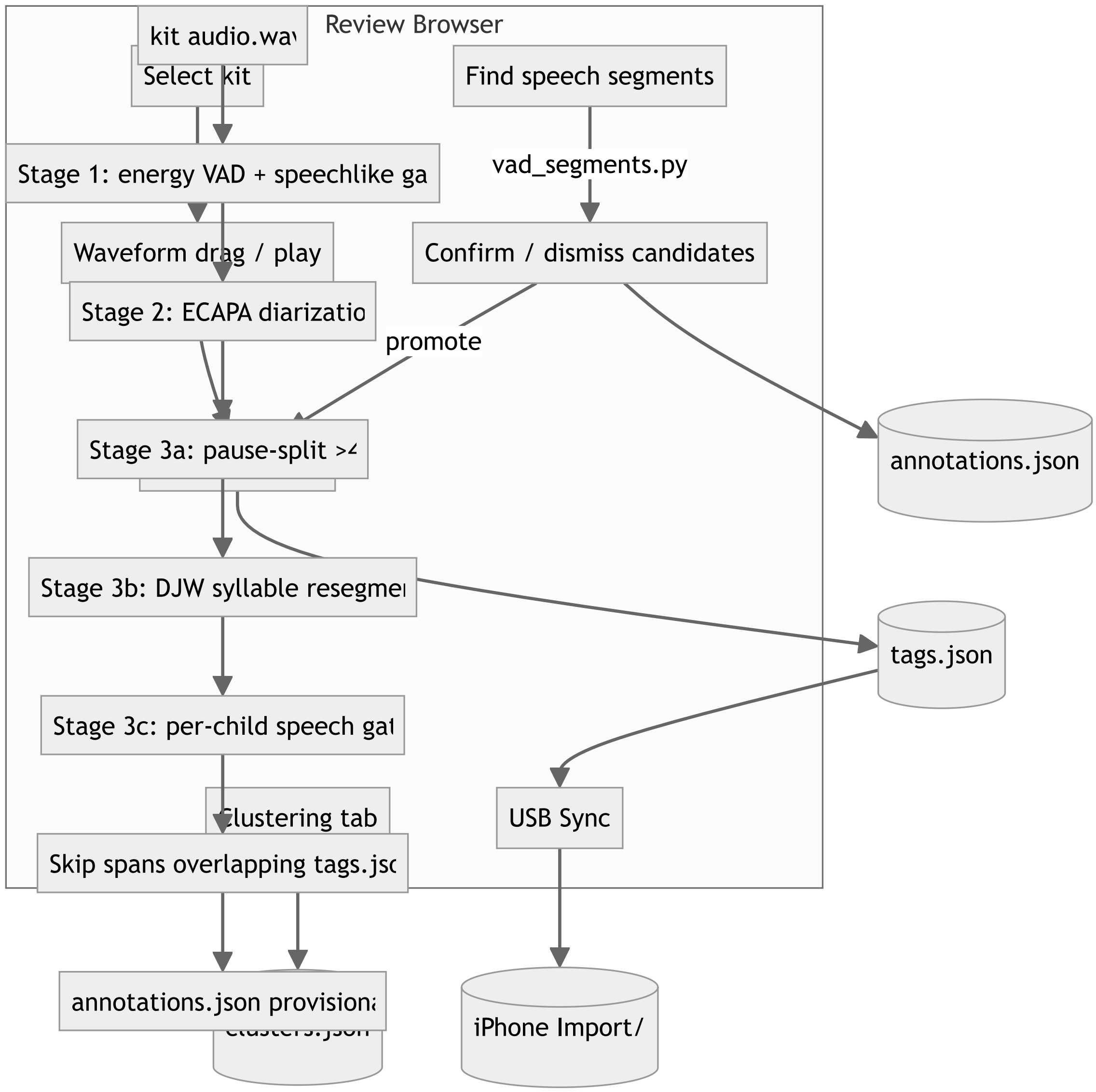
Entry points

Mode	Command	Status
Supervised (preferred)	`tools/run_review_server.sh` → http://127.0.0.1:8765	**WIP**
Direct	`python3 tools/review_server.py`	**main**
Status / stop / restart	`tools/run_review_server.sh --status|--stop|--restart`	**WIP**

Log: `tools/.run/review_server.log` . Kits default to `~/Documents/BabyTalk/Library/` (seeded from `Backups/` on first launch).

UI behaviors

- Live writes to each kit's `tags.json` on Add / Save / Delete
- **Sync with iPhone** — pull kits, push `manifest.json` + `tags.json` only (**main**)
- **Theme:** Auto (evening/night dark) → Light → Dark (**WIP**; `localStorage` key `babytalk-review-theme`)
- **Tags panel:** compact one-line rows; open row expands editor; scroll clamped to ~20 rows (**scroll height WIP**; compact list **main**)
- Tabs: session review (waveform + ML candidates) + **Clustering**
- Optional Whisper **Suggest** on a span (`asr_suggest.py` / faster-whisper) — does not overwrite `word` / `phonetic` until Copy→word



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4. ML candidate pipeline (SoTA path)

Default path (**main**):

CLI:

```
tools/.venv/bin/python tools/vad_segments.py ~/Documents/BabyTalk/Library
tools/.venv/bin/python tools/vad_segments.py <kit> --list-backends
tools/.venv/bin/python tools/vad_segments.py <kit> --no-resegment   # coarser blobs
```

Defaults: merge gaps ≤ 200 ms, drop < 300 ms, dual-threshold edge pad (flat 80 ms + lower-threshold reach ≤ 120 ms). Source `vad_v0`. Re-runs replace still-provisional VAD candidates only; confirmed/dismissed + existing tags are preserved.

4.1 Stage 1 — VAD + speechlike gate (main)

Piece	File	Role
Energy VAD	`vad_segments.py`	"Louder than the room?"
Speech gate	`speechlike.py`	"Vocal-tract-like?" — voicing peak / voiced fraction / speech-band / low-band

Score = `0.40·speech_band + 0.25·voicing_peak + 0.15·voiced_fraction + 0.20·(1 − low_band)`.

Threshold	Behavior
Region screen < **0.42**	Drop whole region before diarization (lenient)
Candidate < **0.55**	Drop
0.55–0.68	Keep, flag `possible_non_speech`, down-rank
≥ **0.68**	Clean pass

Calibration (speech vs noise / dismiss): AUC **~0.81** on 457 reviewer-confirmed (206) / dismissed (251) spans; per-kit 0.77 / 0.83 / 0.88 (`speechlike.py` docstring, `tools/README.md`). Logistic re-fit was worse (LOKO AUC 0.79). Syllabic modulation 2–10 Hz AUC 0.46 — discarded.

"Worth tagging" is a different problem. After the gate, leftover false positives (candidates with no tag overlap on curated fresh kits) are **47.9%, and `speechScore` AUC for hit-vs-miss tags is only** 0.53 (`tools/analysis/out/ml_delta_fresh.json` → `pooledFreshCurated`). Retuning the speechlike threshold will not solve worth-tagging.

4.2 Stage 2 — Diarization (main + VTC branch)

Sliding 1.5 s windows → embeddings → cosine agglomerative clustering → cut on cluster change. Writes `speakerCluster` (`SPEAKER_00`, …) — **not** the reviewer `speaker` field (Confirm still asks).

Backend	Status	Notes
`ecapa`	**main** default	SpeechBrain ECAPA-TDNN; ~80 MB cache; no HF token
`melstats`	**main** fallback	MFCC mean+std; weak (can split one child into many speakers)
`pyannote`	**main** opt-in	Needs `BABYTALK_HF_TOKEN`; never auto-selected
`vtc` / `vtc-first`	**branch**	Role labels KCHI/OCH/FEM/MAL → Baby/Parent/Other; needs local VTC checkout

```
# Keep energy VAD; swap diarizer only
tools/.venv/bin/python tools/vad_segments.py <kit> --diarization vtc

# Skip VAD+ECAPA; use VTC role timeline as parent spans
tools/.venv/bin/python tools/vad_segments.py <kit> --segmentation vtc-first
```

Why VTC failed as stage-2 / vtc-first on these kits

Segment quality (curated fresh, ECAPA default vs `vtc-first`):

Metric	ECAPA path (`ml_delta_fresh.json`)	VTC-first (`ml_delta_vtc.json`)
Tag any-overlap	**92.9%**	41.7%
IoU≥0.5	**50.0%**	17.4%
Missed tags	**7.1%** (14/198)	**58.3%** (134/230)
nCands	315	371

Role accuracy (`role_delta.json` , pooled; n=228 tags with speaker):

System	Baby F1	Adult F1	Notes
VTC (direct roles)	**8.1** (R 4.2%)	**8.7**	~52.6% miss; baby often → Adult/MISS
ECAPA + oracle cluster→class	Baby **96.7**	Adult **0.0** (n=4)	Upper bound if human maps clusters; baby/non-baby macro F1 **89.1**

VTC was trained for LENA-style recorders, not close phone mics in home/hospital rooms. On this library it both **misses baby speech as segments** and **mis-labels roles**. Do not replace ECAPA with VTC for these kits.

4.3 Stage 3 — Pause-split + DJW resegment (main; knob plumbing WIP)

1. Same-speaker spans > **4 s** → deepest relative energy dip; hard cap **15 s** (`hard_capped`)
2. `resegment.py` — de Jong & Wempe (2009) syllable nuclei (Praat intensity + preceding dip + voicing), then BabyTalk word-like cut knobs:

Knob	Default	Role
`WORD_SPLIT_MIN_SEP_MS`	300	Cut if nuclei ≥ this far apart
`WORD_SPLIT_MIN_DIP_DB`	4.0	Or if valley prominence ≥ this
`RESEG_TARGET_MS`	1600	Force-split leftovers still longer
`RESEG_MIN_PART_MS`	420	Minimum child duration

Children carry `parentSpanId`, `resegMethod: dejong_wempe`, `splitBy: syllable`. Disable with `--no-resegment`.

IoU sweep (`tools/analysis/out/iou_sweep.json`, WIP tool):** best config **is the baseline** defaults above. Pooled **manual** IoU≥0.5 = **57.2%** (any overlap 95.2%; nManual=152, nCands=645). Tuning knobs did not beat defaults.

Dominant failure mode: over-split / too-short children (e.g. multi-syllable words like *tea|cher*), not too-long blobs (`candTooLong2xPct` ≈ 0–1%). **Merge-back** (and/or Whisper word timestamps) is the future lever — not tighter splits.

4.4 Clustering tab (main)

`tools/cluster_sounds.py` — log-mel temporal fingerprints + sklearn agglomerative clustering → `clusters.json` (+ optional `cluster_embeddings.npz`). Groups similar *sounds/words* regardless of speaker; labels live on the cluster.

Known limitation: mel summaries still lean on pitch / loudness / duration structure even with light peak normalize — weak for "same word, different prosody."

Contemplated (not built): swap embeddings for HuBERT / wav2vec2; UI workflow **propose clusters → name** (label once, members stay linked for training). Separate from diarization.

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5. Benchmarks & analysis tools

Tool	Status	Purpose
`tools/analysis/ml_delta.py`	**main** (+ **branch** flags)	Tag↔candidate coverage, FP rate, speechScore AUC
`tools/analysis/role_delta.py`	**WIP**	VTC vs ECAPA-oracle role F1
`tools/analysis/iou_sweep.py`	**WIP**	DJW word-split knob sweep vs manual IoU≥0.5

Outputs (examples used for this doc):

- `tools/analysis/out/ml_delta_fresh.json` — ECAPA SoTA fresh curated
- `tools/analysis/out/ml_delta_vtc.json` — `segmentation=vtc-first`
- `tools/analysis/out/role_delta.json`
- `tools/analysis/out/iou_sweep.json`

```
# Fresh ECAPA-path delta (read-only; never writes Library)
tools/.venv/bin/python tools/analysis/ml_delta.py --fresh \
  --out tools/analysis/out/ml_delta_fresh.json

# VTC-first comparison
tools/.venv/bin/python tools/analysis/ml_delta.py --fresh --segmentation
        vtc-first \
  --out tools/analysis/out/ml_delta_vtc.json

tools/.venv/bin/python tools/analysis/role_delta.py \
  --out tools/analysis/out/role_delta.json

tools/.venv/bin/python tools/analysis/iou_sweep.py \
  --out tools/analysis/out/iou_sweep.json
```

Headline numbers (curated kits)

Question	Answer	Source
Speechlike separates speech vs junk?	AUC **~0.81**	`speechlike.py` / README
speechScore ranks "worth tagging"?	AUC **0.53; FP leftover 48%**	`ml_delta_fresh.json`
ECAPA segment recall (any overlap)	**92.9%**	same
VTC-first any overlap	**41.7%**; miss **58%**	`ml_delta_vtc.json`
VTC Baby role F1	**8.1**	`role_delta.json`
ECAPA-oracle Baby F1	**96.7**	same
Best DJW knobs	**defaults**; manual IoU≥0.5 **57.2%**	`iou_sweep.json`

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6. Open priorities (ordered)

1. **Worth-tagging signal** — separate from speechlike. speechScore AUC ~0.53 / ~48% FP means the next filter should target "reviewer would tag this," not another speech-vs-noise retune.
2. **Word-level merge-back** (and/or Whisper timestamps) — fix over-split (*tea*/*cher*); IoU sweep says don't tighten DJW defaults.
3. **Clustering embedding upgrade** — HuBERT/wav2vec2 + propose-and-name UX; keep mel+sklearn until then.

4. **Not VTC-as-ECAPA-replacement** on these phone-mic kits — optional experiment only; default stays ECAPA.

Phone/Mac durability (kit schema, USB sync) is largely in place on **main**; polish continues in Review UI **WIP**.

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7. Data model (session kit)

Library root: `~/Documents/BabyTalk/Library/<kit-folder>/`

File	Role
`manifest.json`	Identity + audio metadata (`recordingUuid` , `sessionName` , `audioFile` , `durationMs` , codec/rate/channels, …)
`audio.wav` (or path in manifest)	Decoded master for Mac tools
`waveform.json`	Envelope for UI — **not** ML input
`tags.json`	Trusted human (and `ml_confirmed`) spans — USB push target
`annotations.json`	Provisional ML candidates (`source: vad_v0` , status provisional/confirmed/dismissed)
`clusters.json`	Acoustic clusters + labels (**optional**, from Clustering tab)
`cluster_embeddings.npz`	Optional embedding cache

Tag / annotation sketch

```
tags.json        { "tags": [ { uuid, startMs, endMs?, category, speaker?,
                              word?, phonetic?, note?, label, source, ... }
] }

annotations.json { "annotations": [ { uuid, startMs, endMs, status, source,
                                     speechScore?, speakerCluster?, flags?,
                                     parentSpanId?, resegMethod?, ... } ] }

clusters.json    { clusters: [ { id, label fields, member ids, confidence… }
] }
```

USB (**main**): pull full kits from phone `Documents/Backups/` → Mac Library; push `manifest.json` + `tags.json` → phone `Documents/Import/<kit>/`. Open BabyTalk on phone to auto-import.

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8. Phone app (brief)

Deep file-tied walkthrough: `ARCHITECTURE_DEEP_DIVE.md`.

Concern	Location
Record / Tag Now / Save + Path B	`src/screens/RecordScreen.tsx`
Playback + seek	`src/screens/PlaybackScreen.tsx`
Waveform envelope	`src/components/Waveform.tsx`, `src/waveform/`
SQLite	`src/db.ts`
Native audio + `extractWaveformPeaks`	`react-native-audio-recorder-player` (local fork)

Path A = live metering; Path B = file-derived peaks after save / on playback load. On-screen bars are a **loudness envelope**, not PCM.

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9. How to run (cheat sheet)

```
# One-time
cd /path/to/babytalkApp-1
python3 -m venv tools/.venv
tools/.venv/bin/pip install -r tools/requirements.txt

# Review UI (WIP launcher preferred)
tools/run_review_server.sh
open http://127.0.0.1:8765
# or: python3 tools/review_server.py

# ML candidates (ECAPA SoTA)
tools/.venv/bin/python tools/vad_segments.py ~/Documents/BabyTalk/Library

# USB
tools/.venv/bin/python tools/iphone_sync.py sync

# Analysis (read-only)
tools/.venv/bin/python tools/analysis/ml_delta.py --fresh --out
        tools/analysis/out/ml_delta_fresh.json
tools/.venv/bin/python tools/analysis/role_delta.py --out
        tools/analysis/out/role_delta.json   # WIP
tools/.venv/bin/python tools/analysis/iou_sweep.py
        # WIP
```

Phone: see root `README.md` / `INSTALL_TO_IPHONE.md`.

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10. File map (Mac tools)

```
tools/
├── review_server.py          # Review Browser (HTML+API)
├── run_review_server.sh      # Supervised launcher (WIP)
├── iphone_sync.py            # USB pull/push
├── vad_segments.py           # Pipeline orchestrator
├── speechlike.py             # Absolute speech gate
├── diarize.py                # ECAPA / melstats / pyannote
├── vtc.py                    # VTC roles (branch)
├── resegment.py              # DJW syllable cuts
├── cluster_sounds.py         # Mel + sklearn clustering
├── asr_suggest.py            # Optional Whisper
├── propose_candidates.py     # Legacy short-burst onsets
├── validate_export.py
├── requirements.txt
└── analysis/
    ├── ml_delta.py
    ├── role_delta.py         # WIP
    ├── iou_sweep.py          # WIP
    └── out/*.json            # Benchmark dumps (WIP / local)
```

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11. Gaps / verification notes

- **HuBERT / propose-and-name:** product direction only; no implementation in tree yet.
- **tea|cher example:** failure-mode class inferred from IoU sweep (too-short / over-split, low `candTooLong2xPct`); not a single logged clip id in JSON.
- **Dark theme / Tags viewport / `run_review_server.sh` / `role_delta` / `iou_sweep`:** verified in worktree; confirm before treating as shipped.
- **`ml_delta_fresh.json` `segmentation` / `diarization` fields** were `null` in the dump header while metrics match the ECAPA SoTA path — treat coverage numbers as authoritative; re-run with explicit flags if you need stamped metadata.
- Phone export kit phases in older roadmap text are largely **done on main**; `ARCHITECTURE_DEEP_DIVE.md` §12 still describes some as “planned” — prefer this doc + `tools/README.md` for Mac/ML currency.
- Adult role support in `role_delta` is tiny (n=4 Adult tags) — Adult F1 comparisons are weak; Baby metrics dominate.